PIYUSH KUMAR

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SUMMARY

Software engineer specializing in Applied AI/GenAI application development, built on a full-stack engineering foundation (React, Node.js, PostgreSQL). Designed and built a retrieval-augmented generation (RAG) system that grounds LLM responses in retrieved content covering transcript chunking, embedding generation, vector retrieval with pgvector, and LangChain orchestrated generation. Combines LLM application architecture with production web engineering: real-time systems, authentication, and database design.

EDUCATION

Bachelor of Engineering (B.E.) - Computer Science & Engineering
Visvesvaraya Technological University, Bangalore · CGPA: 9.2/10.0 · Dec 2022 -Jun 2026

EXPERIENCE

Software Development Intern - Mindmatrix, Bangalore
Jan 2026 - May 2026

- Built GenAI powered product features, including AI assisted conversational interfaces and content generation flows, integrated into the core product experience.
- Shipped production ready UI features using component based architecture within agile sprint cycles, under mentor review.
- Contributed to sprint planning, peer code review, and technical documentation as part of a cross functional engineering team.

PROJECTS

LearnLens - AI Video Learning Assistant
LangChain · OpenAI Embeddings API · PostgreSQL/pgvector · Semantic Retrieval

- Built a retrieval augmented generation (RAG) pipeline that grounds LLM responses in actual video content, transcripts are chunked, embedded via the OpenAI embeddings API, and indexed in PostgreSQL with pgvector for semantic retrieval replacing full transcript prompting with targeted, relevant context.
- Orchestrated the retrieval, prompt-construction, generation flow using LangChain, so summaries and answers are built from the transcript chunks most relevant to each query rather than the entire transcript.
- Designed follow-up Q&A with session-level conversational memory, enabling multi-turn questions about a video where each answer stays grounded in retrieved content and prior conversation context.
- Extended the retrieval pipeline to generate topic-based quizzes from retrieved transcript segments using structured LLM output.

DevLink - Real-Time Developer Networking Platform
Node.js · Express.js · Socket.io · PostgreSQL · React · JWT

- Architected real-time features using Socket.io for bi-directional communication, supporting live messaging and activity feeds.
- Implemented JWT based authentication with refresh token rotation and hardened session handling.
- Built full-text profile search using PostgreSQL tsvector indices and API rate limiting middleware.
- Designed normalized PostgreSQL schemas to model developer connection graphs and profile data.

GlitchPop - React UI Component Library
React · TypeScript · Tailwind CSS · Radix UI · Shadcn UI

- Built accessible, production-ready UI components (modals, forms, tabs, buttons) adhering to WCAG 2.1 standards, using a fully-typed TypeScript architecture with design tokens for theming.
- Centralized previously duplicated UI logic into a single reusable library layered on Radix UI primitives and Tailwind CSS, reducing duplication across consuming projects.

SKILLS

AI / GenAI: LLM, RAG, LangGraph, LangChain, OpenAI API, Prompt Engineering
Backend: Node.js, Express.js, REST APIs, WebSockets, Microservices
Languages: C++, Java, JavaScript, Python
Frontend: React.js, TypeScript, HTML5, CSS3, Tailwind CSS
Databases: PostgreSQL, MongoDB, MySQL, SQL
Tools: Git, GitHub, Docker, AWS CLI, Postman, Linux